

Does a Carbon Tax Reduce CO₂ Emissions?

Evidence from British Columbia.

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ECON 712: Applied Econometrics
Replication Exercise (Felix Prestis 2022)

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April 20, 2026



Outline

- ① Summary and contribution
- ② Identification strategy
- ③ Data and results
- ④ Assessment and improvements



Summary and Contribution

► Summary

- Studies the effect of the 2008 BC carbon tax on CO₂ emissions ([Pretis 2022](#))
- Uses a difference-in-differences framework with dynamic adjustment
- Finds no significant effect on aggregate emissions, but a significant reduction in transportation emissions

► Research question

- Does the 2008 BC carbon tax reduce CO₂ emissions? ([Pretis 2022](#))

► Motivation

- Carbon taxes are a key policy tool for reducing emissions ([Carattini et al. 2019](#))
- Existing evidence shows stronger effects in transportation than aggregate emissions ([Andersson 2019](#), [Davis & Kilian 2011](#))
- It is unclear whether sectoral reductions translate into overall emission declines

► Contribution

- Places emphasis on aggregate emissions rather than sector-specific outcomes.

Methodological contribution

- Combines DiD, synthetic control, and break detection methods, identifies other structural changes ([Castle et al. 2015](#))
- Uses randomization inference for a single treated unit, improves statistical credibility of the results ([Cameron et al. 2008](#))

Limitation

- Identification relies on parallel trends assumptions.
- A single treated unit, which may bias estimates and reduce statistical precision

► Findings

- No statistically significant effect of the carbon tax on aggregate CO₂ emissions
- Significant reduction in transportation emissions consistent with ([Andersson 2019](#), [Metcalf & Stock 2020](#))
- Limited or no response in other sectors (e.g., stationary energy, industry)

Identification Strategy

- Uses a difference-in-differences (DiD) framework ([Angrist & Pischke 2009](#))
- Compares changes in emissions in BC (treated) to other provinces (control) before and after 2008
- Treatment defined as $Tax_{i,t} = Treat_i \times Post_t$
- Key assumption: parallel trends in the absence of the policy
- Includes province (μ_i) and year (λ_t) fixed effects to control for unobserved heterogeneity ([Pretis 2022](#))
- Uses randomization inference to address inference with a single treated unit ([Cameron et al. 2008](#), [Pretis 2022](#))

► The Empirical Model

$$\log(Emiss)_{i,t} = \lambda_t + \mu_i + \beta Tax_{i,t} + \alpha \log(Emiss)_{i,t-1} + \gamma_1 \log(Pop)_{i,t} + \gamma_2 \log(GDP)_{i,t} + \varepsilon_{i,t}$$

- $\log(Emiss)_{i,t}$: log of CO₂ emissions in province i at time t
- λ_t : year fixed effects
- μ_i : province fixed effects
- $Tax_{i,t}$: treatment indicator, equal to $Treat_i \times Post_t$
- β : short-run effect of the carbon tax
- $\log(Emiss)_{i,t-1}$: lagged emissions, capturing persistence over time
- $\log(Pop)_{i,t}$: log of population
- $\log(GDP)_{i,t}$: log of real GDP
- $\varepsilon_{i,t}$: error term



► Dynamic Setting

- The dynamic setting captures how emissions adjust over time, so the equilibrium effect reflects the full long-run impact of the carbon tax

$$\beta^{eqbm} = \frac{\beta}{1 - \alpha}$$

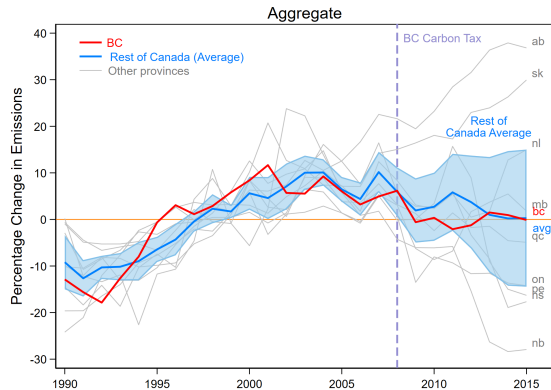
- β : short-run effect of the carbon tax
- α : persistence parameter from lagged emissions
- β^{eqbm} : long-run (equilibrium) effect after full adjustment
- Captures total impact accounting for dynamic responses over time
- Standard errors computed using the delta method

► Data

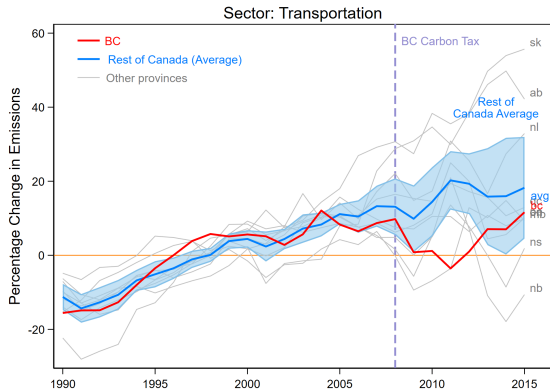
- Panel dataset of Canadian provinces 1990-2023, BC treated
 - Used only 10 Canadian Provinces and excluded territories
 - Only CO₂ emission figures were estimated
 - Construct treatment variable: $Tax_{i,t} = Treat_i \times Post_t$
 - Sector-level emissions classified using Intergovernmental Panel on Climate Change (IPCC) classifications (Energy, Fugitive, Industrial Processes, Transportation, Agriculture, and Waste)
 - Control variables were Population and GDP from Statistics Canada
- Population controls for changes in the scale of economic activity and energy demand
- GDP controls for economic activity and income-driven emissions

Identification Strategy cont'd

► Descriptive Evidence: Pre-trends/Parallel trends graph



Aggregate emissions



Transportation emissions

Main Results

- Coefficients are generally negative, suggesting the carbon tax reduces emissions, but effects are mostly statistically insignificant
 - Total emissions: small and insignificant effect ($\beta = -0.0071$), implying no clear impact on aggregate emissions
 - Transportation: statistically significant reduction at the 1% level ($\beta = -0.0223^{***}$), corresponding to about a 2.2% decrease in emissions
 - Other sectors: effects are negative but statistically insignificant, indicating limited or imprecise responses outside transportation
- ⇒ Transportation: strong long-run reduction ($\beta^{eqbm} = -0.1429^{***}$), implying about a 14.3% decline, reinforcing the significant short-run effect

Main Results Cont'd

	<i>Dependent variable: $\log(\text{Emissions})_{it}$</i>						
	Total	Stat. Energy	Transportation	Fugit. (excmp)	Industry	Agriculture	Waste
Carbon Tax _{it}	-0.0071 (0.0056)	-0.0040 (0.0055)	-0.0223*** (0.0066)	-0.0673 (0.0652)	-0.0882 (0.0572)	-0.1590 (0.1259)	–
$\log(\text{Emiss})_{i,t-1}$	0.887*** (0.0386)	0.880*** (0.0392)	0.844*** (0.0487)	0.962*** (0.0194)	0.714*** (0.0741)	0.687*** (0.1182)	0.921*** (0.0455)
$\log(\text{Pop})_{it}$	-0.007 (0.0417)	-0.008 (0.0489)	-0.095** (0.0335)	0.612** (0.2558)	0.746** (0.2800)	-0.785 (0.9035)	0.458 (0.3304)
$\log(\text{GDP})_{it}$	0.226** (0.0764)	0.229** (0.0792)	0.314*** (0.0898)	0.230 (0.7590)	0.121 (0.3643)	1.688 (1.2321)	-0.243 (0.3009)
Eqbm. Tax Coef	-0.0630	-0.0331	-0.1429***	-1.7726	-0.3087*	-0.5082**	–
Eqbm. Se	(0.0459)	(0.0438)	(0.0321)	(2.2388)	(0.1728)	(0.2396)	
Pct. Impact	-0.7%	-0.4%	-2.2%***	-6.5%	-8.4%	-14.7%	–
Pct. Impact Se	(0.6%)	(0.6%)	(0.7%)	(6.1%)	(5.2%)	(10.7%)	
Pct. Eqbm. Impact	-6.1%	-3.3%	-13.3%***	-83.0%	-26.6%*	-39.8%**	–
Pct. Eqbm. Impact Se	(4.3%)	(4.2%)	(2.8%)	(38.0%)	(12.7%)	(14.4%)	
Observations	330	330	330	297	330	323	170
Clusters	10	10	10	9	10	10	7
R^2 (within)	0.860	0.852	0.840	0.846	0.601	0.614	0.831
F Statistic	562.46***	741.75***	253.45***	1724.97***	118.94***	349.28***	9362.85***

Province and year fixed effects included. Standard errors clustered at the province level (in parentheses). Equilibrium impact equals $\beta/(1 - \rho)$. Percent impacts are computed as $100(e^\beta - 1)$. Percent impact standard errors are obtained using the delta method transformation. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.



► Strength

- CO₂ emissions are an appropriate outcome for evaluating climate policy
- Sectoral analysis reveals important heterogeneity across emission sources
- GDP and population controls capture macroeconomic conditions and scale differences
- Log-linear specification allows clear percentage interpretation of effects
- Inclusion of lagged emissions captures persistence in emissions dynamics
- Use of randomization inference improves validity of statistical inference ([Cameron et al. 2008](#))

► Limitation

- Small number of provinces and a single treated unit limits statistical power
- Clustering at the province level may be unreliable with few clusters ([Cameron et al. 2008](#))
- Potential omitted variables like energy prices

Proposed Improvements

- Allow for heterogeneous treatment effects across provinces (e.g., by industrial structure or energy intensity)
- Strengthen identification using methods suited for a single treated unit (e.g., synthetic control)
- Compare results across alternative specifications to improve robustness
- Use more disaggregated data (firm or plant level) to capture within-province variation

Conclusion

- The carbon tax led to reductions in transportation emissions but not in total emissions
 - Suggests uneven effects across sectors and limited aggregate impact
 - Identification strategy is reasonable but relies on strong assumptions with a single treated unit
 - The results are informative, but their interpretation depends on the validity of the underlying identification assumptions.
- ⇒ Key takeaway: sectoral responses do not necessarily translate into aggregate emission reductions

Thank you

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